

Oscar Antonio Juarez Jimenez

Artificial Intelligence Engineer

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I am an Artificial Intelligence Engineer with experience as a Data Engineer, specializing in the design and development of large-scale data processing pipelines. I am driven to build robust, data-driven solutions that optimize processes and extract actionable value from information. I am deeply passionate about solving complex problems, mastering emerging technologies, and contributing to high-impact projects by bridging logic, analytics, and creativity.

Education

National Polytechnic Institute (IPN)- Escuela Superior de Computo

México, CDMX. Aug 2021- Jul 2025

Bachelor's Degree in Artificial Intelligence Engineering

GPA: 9.5/10

Relevant Courses: Machine Learning, Digital Image Processing, Natural Language Technologies, Neural Networks and Deep Learning, Advanced Neural Networks, Software Engineering for Intelligent Systems, Parallel Computing, Bio-Inspired Algorithms.

Professional Experience

Data Engineer | Bluetab - BBVA

Jul 2025 – Present

- Implemented data ingestion, transformation, and storage (ETL) processes, seamlessly integrating multiple sources into the bank's core data infrastructure.
- Managed datasets and tables in AWS S3, ensuring highly structured storage and efficient access for Spark processing and advanced data analytics.
- Maintained and optimized data pipelines in distributed computing environments (clusters), guaranteeing data reliability, quality, and high availability.
- Collaborated on data integration strategies for analytical use cases and Machine Learning models within the structural risk division, fostering the development of data-driven solutions across the banking ecosystem.

Innovation and Development Intern | Grupo Salinas

Feb – Jul 2025

- Contributed to the research, development, and deployment of artificial intelligence tools aimed at optimizing business workflows and validating models in innovation-driven environments.
- Participated in the engineering of AI agents, conversational chatbots, visual prototypes, and dynamic landing pages.

Banco de México | Full-Stack Intern

Jun - Dec 2024

- Boosted system performance by 30% by optimizing SQL queries and reducing algorithmic complexity in Java/JavaScript, while enhancing UX through a streamlined, intuitive interface.
 - Optimized SQL database management by enforcing integrity and security via normalization, stored procedures, views, indexing, and role-based access control.
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Projects

Object Segmentation System for Indoor 3D Model Editing Using NeRF

Jun 2024 – Aug 2025

- Designed an AI-based object segmentation system for editing 3D models in interior environments.
- Boosted system classification accuracy by 30% by integrating Machine Learning algorithms into the initial U-Net architecture.
- Achieved photorealistic 3D editing by combining NeRF's spatial reconstruction with diffusion-based texture synthesis (Stable Diffusion).

RAG ChatBot

Jan – Jun 2024

- I collaboratively developed a Retrieval-Augmented Generation (RAG) system for LLM to generate summaries and answer natural language queries about Mexico's Diario Oficial de la Federación (DOF).
- Used LangServe and LangChain for model retrieval and Pinecone for vector storage.

To check out more of my projects, please feel free to visit my own portfolio at <https://oscarjuji.github.io/>

Courses & Certifications

- Legacy Responsive Web Design Challenges, Legacy Python for Everybody & JavaScript Algorithms and Data Structures – FreeCodeCamp

- Learn AI Skills Challenge (Azure) – Microsoft
 - Google Cloud Computing Foundations Certificate
 - Google Cloud GenAI Skill Badge Pathways
 - Oracle Next Education - Alura LATAM
 - Secure Development 3.0 – Net4skills
 - Jira Scrum Team
 - AWS Technical Essentials
 - SAFe Scrum Master Exam by Scaled Agile
 - AWS Certified Data Engineer – Associate (*In Progress)
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Skills

Programming Languages: Intermediate (2 years): C/C++, Python, Java, JavaScript, SQL, and Excel; Basic (≤ 1 year): C#, CUDA, Go, PHP, TypeScript, R, Scala.

Tools & Technologies: Intermediate (2 years): Git, Google Cloud Platform (GCP), Docker, HTML/CSS, AI/Machine Learning, Tableau; Basic (1 year): Power BI, MLflow, Kubernetes, Azure, Amazon Web Services (AWS).